1 Induction to prove alternating sign on implies roots

Statement: $Q(n)$ if every adjecent pair of integers in the set $\{0,\ldots,n\}$, when $n \geq 1$, alternate sign under f (e.g. $f(0) < 0 \wedge f(1) > 0$), then there is a set S that contains root of f, where $|S| = n$. Formaly

$$Q(n)\colon \forall l \in \{0,\ldots,n\}\colon R(f,l) \implies \exists S\Big((|S| = n) \wedge (\forall s \in S : f(s) = 0)\Big)$$

Base Case: $n = 1$

$$Q(1)\colon \forall l \in \{0,1\}\colon R(f,l) \implies \exists S\Big((|S| = 1) \wedge (\forall s \in S : f(s) = 0)\Big)$$

Assume that for some f, the image of 0 and 1 under f is $f(0) \leq -\frac{1}{6}$ and $f(1) \geq \frac{1}{6}$ respctively. Then, by Intermediate value theorem, theres some $s_0 \in [0,1]$ satisfying $f(s_0) = 0$.

Take $S = \{s_0\}$. Clearly, $|S| = 1$, which is what we wanted. This completes the base case.

Inductive case: Supppose Q (k-1) is true for $k - 1 \geq 2$. Then we must show it is also true $Q(k+1)$.

Use the inductive hypothesis to obtain the S_{k-1} where S satisfies

$$\Big((|S| = k-1) \wedge (\forall s \in S : f(s) = 0)\Big)$$

We want to show there is a set S_k, where $S_{k-1} \subsetneq S_k$ and $S_{k-1} \cap S_k =$, and where S_k satisfies $\Big((|S| = k) \wedge (\forall s \in S : f(s) = 0)\Big)$.

Case 1: $k-1$ is even; moreover, by assumption $P(f, k-1)$ is true. This means that $k - 1 \geq \frac{1}{6}$.

Simularly, because $k-1$ is even, k is odd. by assumption $P(f,k)$ is true. This means that $k \leq -\frac{1}{6}$.

Therefore, we have two points for which the range of f cross from negative to positive. By intermediate value thoerem, there must be a root, call it s_k

Observe that $\{s_k\} \cap S_{k-1} =$, since the intervals, $[0,1]\ldots[k-2,k-1]$, which produced S_{k-1} are strictly less then k.

Therefore take $S_k = \{s_k\} \cup S_{k-1}$.

Finally, note that for any two disjoint sets A, B, we have that $|A \cup B| = |A| + |B|$. So, $|S_k| = |\{s_k\} \cup S_{k-1}| = 1$ + k - 1.

This gives the claim, in the case that k is even.

Case 2: By similar argumentation made in case 1, this case also is true.

This completes the inductive hypothesis.

Conclusion: We have shown the claim to be true.